MILITARY SPECIFICATION

REGULATOR, AIR, DEMAND, DIVER'S

1. SCOPE

1.1 This specification covers a demand type regulator for use with self-contained underwater breathing apparatus that will provide a diver with sufficient air on inhalation and will stop flow of air at the end of inhalation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.

PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 - Boxes, Wood, Cleated-Plywood.

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 - Boxes, Fiber.

PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116 - Preservation, Methods of.

MIL-D-963 - Drawings, Production, Electrical Hull and Mechanical Equipment.

MIL-B-10377 - Boxes: Wood-Cleated, Veneer, Paper Overlaid.

MIL-L-10547 - Liners, Case, Waterproof.

MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).

MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS

S7603-1, 543, 156 - Demand Apparatus Breathing Tube.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction unit. - Before production is started, a regulator shall be submitted for testing in accordance with 4.2.

3.2 Material. - All material used shall be non-magnetic and, except as specifically authorized in

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this specification, shall be made from appropriate metals suitable for the purpose intended.

3.3 The demand regulator may be either a single or two stage type.

3.4 The demand regulator shall operate in accordance with all requirements of this specification when used with high pressure air flasks. When using high pressure cylinders the apparatus shall operate satisfactorily in accordance with this specification when swimming to 200 feet.

3.5 When used with high pressure cylinders the demand regulator shall function in accordance with this specification from 3000 pounds per square inch (p.s.i.) to 100 p.s.i.

3.6 The demand regulator shall be designed so that regardless of the swimmer's position the regulator will not give continuous flow.

3.7 The demand regulator shall be compatible with a manifold consisting of a plug in each cylinder leading to a common yoke type valve. It shall be mounted directly to the cylinder manifold valve through a yoke type fitting.

3.8 All parts subjected to high pressure shall be made of brass. The metal parts shall be coated to give maximum resistance to corrosion and inhibit collection of vertigris.

3.9 The complete demand regulator shall have a nonmagnetic permeability of not more than 1.004.

3.10 In inlet and exhaust breathing, tubes shall be of proper size to connect to a 1- inch inside diameter (i.d.) corrugated hose shown in Drawing S7603-1, 543, 156. There shall be a bead on the regulator inlet and exhaust tubes. The bead shall fit into the first corrugation of the breathing tube.

3.11 The air inlet of the demand regulator or first stage regulator shall have a cindered bronze filter. The filter porosity shall be determined on the basis of the valve opening of the demand regulator.

3.12 The exhaust valve shall be located so that the bubbles will not interfere with diver's vision regardless of his position in any direction.

3.13 The demand regulator or first stage and demand regulator combination shall pass not less than the quantities of gas shown on figure 1 when tested in accordance with 4.5.2.

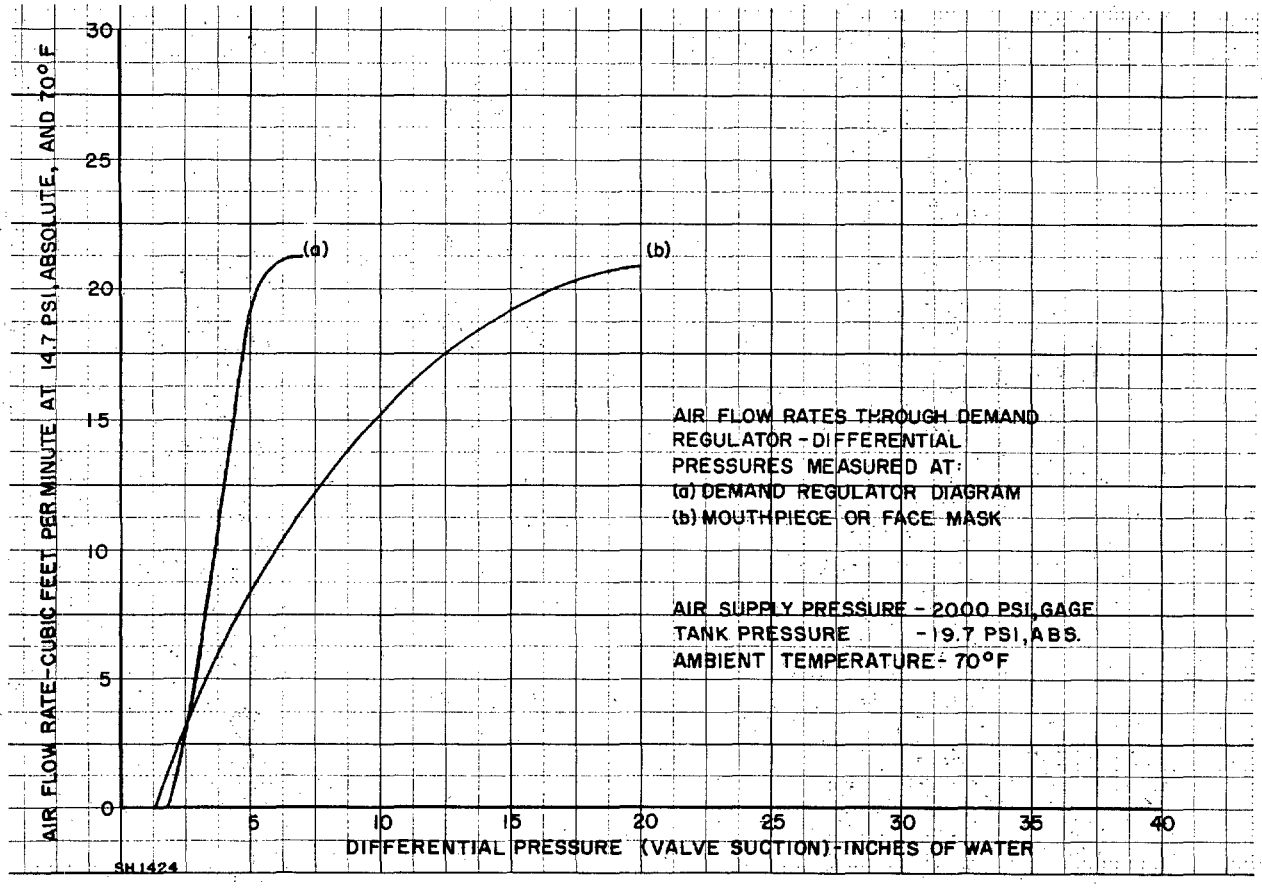


FIGURE 1.

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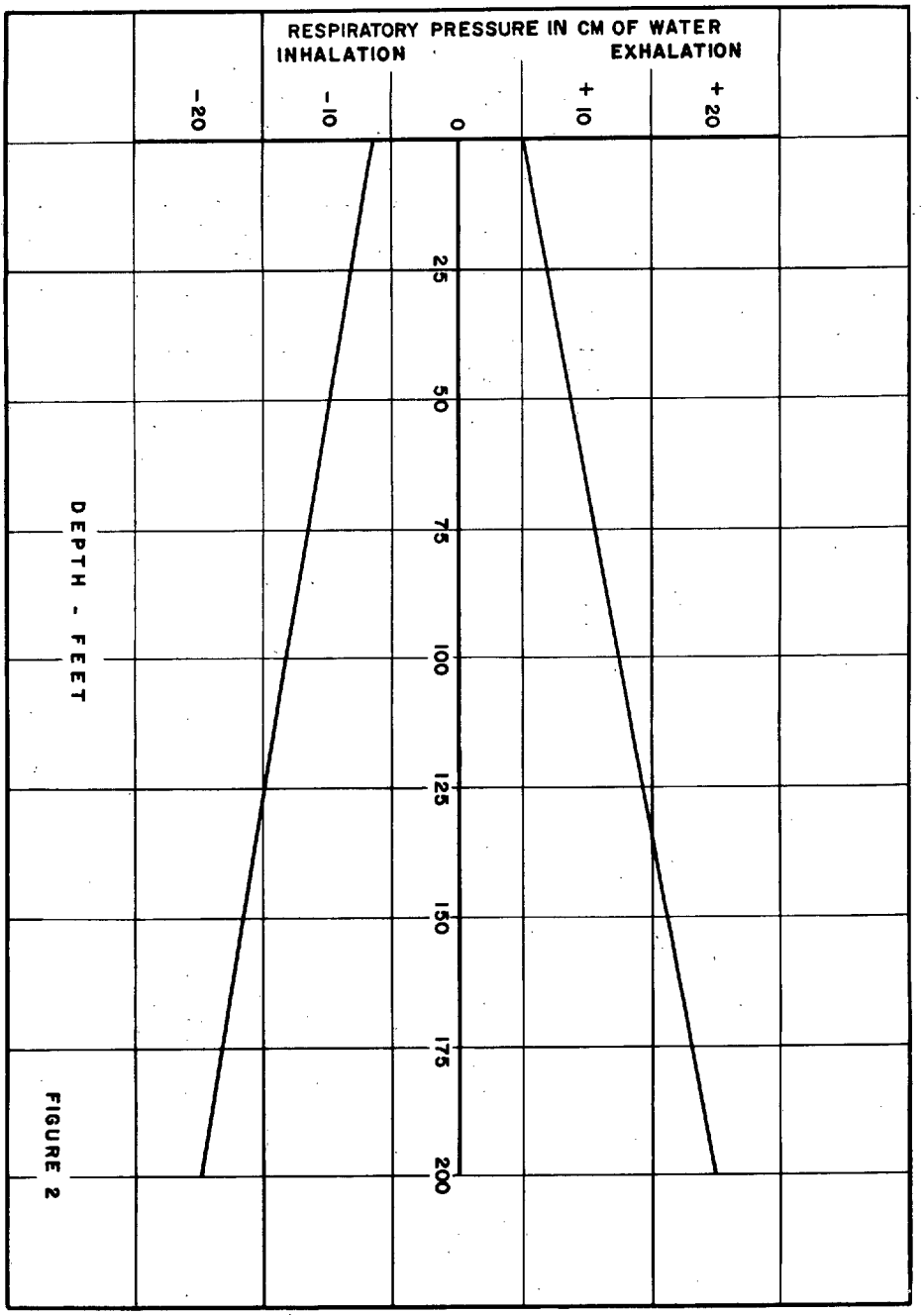


FIGURE 2

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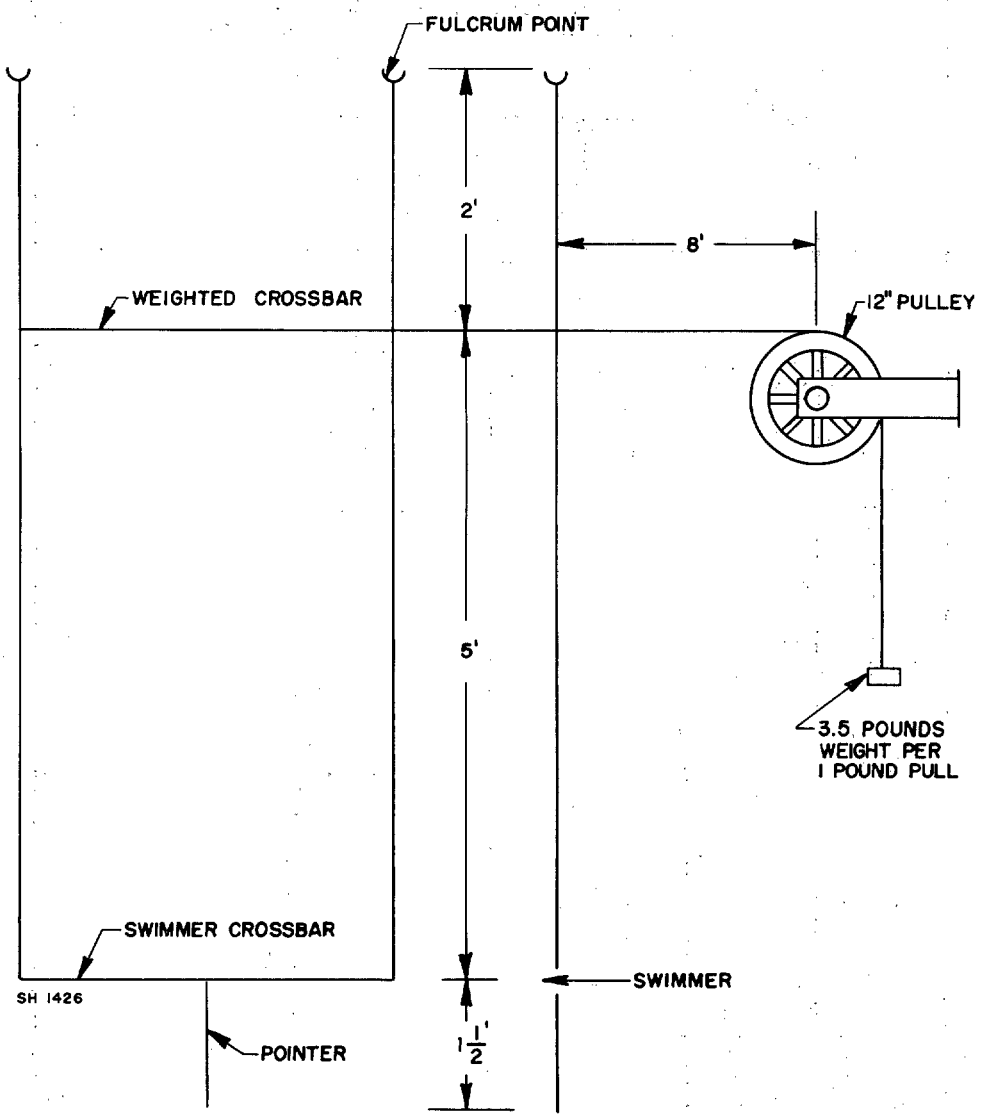
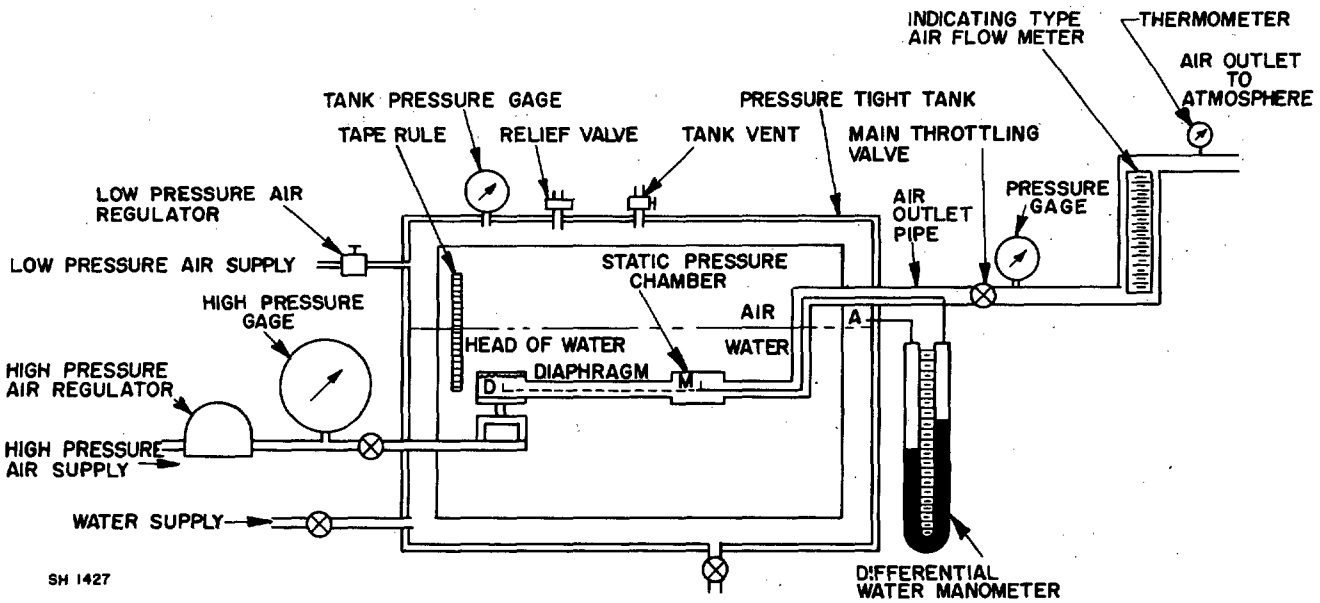


FIGURE 3.

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A - PRESSURE TAP AT TANK TOP AIR SPACE.
D - PRESSURE TAP AT DEMAND VALVE DIAPHRAGM.
M - PRESSURE TAP AT DEMAND ASSEMBLY MOUTHPIECE OR FACE MASK.

FIGURE 4.

3.14 The demand regulator in combination with breathing tubes, mouthpiece, or face mask shall have satisfactory inhalation and exhalation pressures with a man swimming at speed of 1 mile per hour and have a maximum tidal volume of 5 liters. The inhalation and exhalation pressures shall not exceed the values shown on figure 2 for machine test with breathing tubes and mouthpiece or with mask when tested as specified in 4.2.

3.15 The demand regulator in combination with breathing tubes and mouthpiece or face mask shall not have inhalation or exhalation pressures which exceed 200 centimeters of water at 200 feet when swimming against a 12-pound force (see 4.2).

3.16 There shall be a plastic or other suitable protective cap for the air ports.

3.17 Flow rate. - When tested in accordance with 4.5.2, the total differential pressure, or suction, on the sample valve, required to produce the specified flow rate, shall not exceed 5 inches of water. When tested in accordance with 4.5.4, the differential pressure resulting from the specified flow rate shall not exceed 4 inches of water.

3.18 Supply pressure. - When tested in accordance with 4.5.3, the air flow rate at reduced pressure shall be not less than 60 percent of the air flow rate at the initial air supply pressure.

3.19 Repair parts. - The bidder shall submit a list of recommended repair parts for approval of the bureau or agency concerned. The repair parts shall be in accordance with Specification MIL-P-15137.

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3.20 Special tools.- The design of the regulator shall be such that the need of special tools for adjustment and maintenance shall be kept to a minimum. Special tools shall be supplied by the contractor. A list of special tools shall be supplied to the bureau or agency concerned for approval. Special tools are defined as those tools not listed in the Catalog of Naval Material, General Stores Section. (Copies of this catalog may be consulted in the office of the Government inspector.)

3.21 Drawings.- Drawings shall be in accordance with Specification MIL-D-963 and shall be furnished to the bureau or agency concerned.

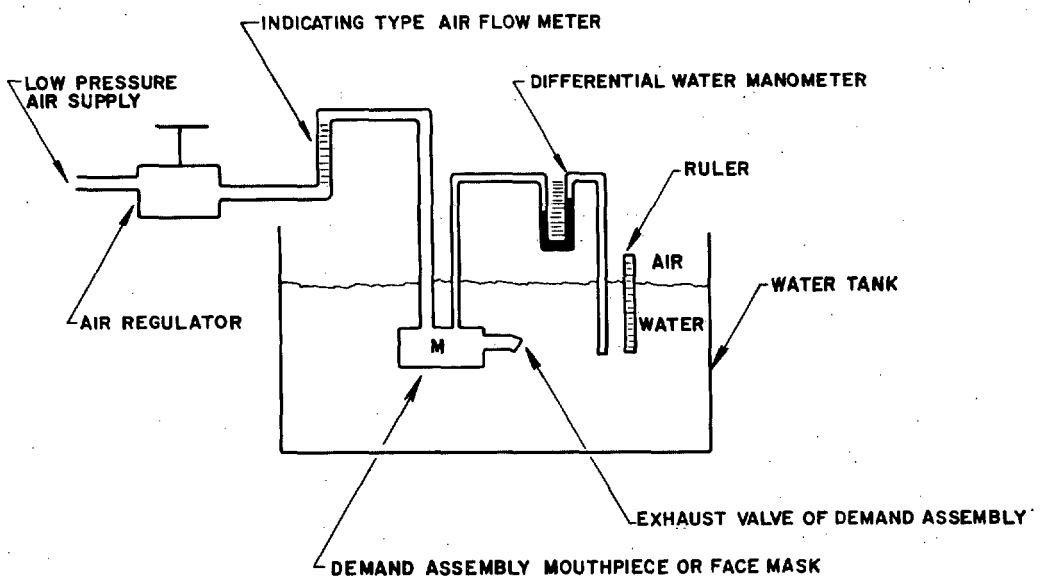
3.22 Technical manuals.- Two complete technical manuals in accordance with type II of Specification MIL-M-15071 shall be furnished with each demand regulator.

3.23 Workmanship.- Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests.-



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FIGURE - 5.

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4.2.1 Contractor.- Prior to submission of the preproduction unit for testing in accordance with 4.2.2, the unit shall be subjected to the tests specified in 4.5.

4.2.2 Government.- After completion of testing in accordance with 4.2.1, the unit shall be subjected to the following tests at the Experimental Diving Unit of the Naval Weapons Plant:

(a) The first test shall be a simulated breathing test using a breathing machine having a tidal volume of 2 liters and set at a rate of 20 breaths per minute.
(b) The second test shall be simulated swimming runs using the arrangement shown on figure 3 and swimming against a 12-pound force.

4.3 Sampling.-

4.3.1 Lot.- All regulators offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for visual and dimensional examination and flow rate test.- From each lot of regulators, samples shall be selected for examination in accordance with Standard MIL-STD-105 at inspection level III for examination of 4.3 and the flow rate test of 4.5.2.

4.3.3 Sampling for operating tests.- From each lot of regulators, samples shall be selected for testing in accordance with inspection level L-8 of Standard MIL-STD-105 and subjected tothe tests specified in 4.5. If any regulator fails in any test, the lot shall be rejected.

4.4 Examination.- Each of the sample regulators selected in accordance with 4.3.2 shall be examined to verify conformance with all of the requirements of this specification which do not involve tests. Any regulator having one or more defects shall be rejected and lots shall be accepted or rejected in accordance with acceptable quality level (A.Q.L.) = 1.5 percent.

4.5 Test procedures.-

4.5.1 General.- The sample demand valve and regulator or complete demand assembly shall be mounted in the test tank and shall be connected to the appropriate inlet and outlet air lines as shown on figure 4. The throttle valve in outlet air line shall be closed and high pressure air, at a pressure of 3000 plus or minus 25 pounds per square inch gage (p.s.i.g.), shall then be admitted to the demand valve regulator. The tank shall then be partially filled with water up to a level such that the assembly under test is submerged to depth of 2 inches above the top of the sample. The sample under test shall then be examined for signs of air leakage and made completely airtight prior to test. Low pressure air shall then be admitted into the top of the tankand the pressure in the air space at the tank top shall be raised to, and maintained at 5 plus or minus 0.2 p.s.i.g. for all tests. All tests shall be conducted at an ambient temperature of 70° plus or minus 5° F.

4.5.2 Flow rate.- The throttle valve in the air outlet line shall be opened slowly and shall be adjusted to a setting such that the air flow rate through the air outlet line, as indicated on the direct reading flowmeter, is 12 plus or minus 0.5 cubic feet of free air per minute for tests of demand valves and regulators only, or 6 plus or minus 0.25 cubic feet of free air per minute for tests of complete demand assemblies. Under steady flow conditions, readings shall be taken of the differential pressure, as indicated by the water manometer, and of the head of water above the center of the demand valve diaphragm, as indicated by the tape rule mounted inside the tank. The sum of these pressures, in inches of water, shall be the differential pressure, or suction, on the sample under test.

4.5.3 Supply pressure.- The throttle valve in the air outlet line shall be opened slowly and shall be adjusted to a setting such that the air flow rate through the air outlet line, as indicated on the direct reading flowmeter, is 8 plus or minus 0.5 cubic feet of free air per minute. When steady flow conditions have been obtained, the total differential pressure, or suction on the sample shall be noted. With this valve of suction maintained constant by suitable adjustment of the throttle valve in the air outlet line, the air supply pressure to the sample valve or assembly shall be reduced from 2000 p.s.i.g. to 300 plus or minus 10 p.s.i.g. and a reading shall be taken of the air flow rate at this air supply pressure, when steady flow conditions have again been obtained. This air flow rate shall be the air flow rate at reduced pressure. Samples equipped with air reserve valves shall have the air reserve valve in the open position for this test.

4.5.4 Exhaust pressure test.- The exhaust assembly including the mouthpiece or face mask and all exhalation tubes and check valves or flapper valves shall be mounted in a test tank and connected to a supply of low pressure air as shown on figure 5. The tank shall be partially filled with water to a depth of 2 inches above the last exhaust valve. One side of a differential water manometer shall be connected to a tap in the mouthpiece or face mask and the other side shall be connected to a tube which has its open end at the same level as the final exhaust valve. The air regulator shall be adjusted to produce a flow of air through the exhaust assembly of 6 plus or minus 0.25 cubic feet of air per minute.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1).-

5.1.1 Level A.- Regulators shall be packaged in accordance with method III of Specification MIL-P-116. Contact preservative shall not be used. The inlet and exhaust breathing tube openings shall be sealed with plugs and pressure sensitive tape.

5.1.2 Level C.- Regulators shall be preserved and packaged in accordance with the supplier's commercial practice.

5.2 Packing (see 6.2).-

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5.2.1 Level A.- Regulators, packaged as specified, shall be packed in overseas type wood cleated fiberboard, nailed wood, fiber, wirebound wood, wood cleated veneer paper overlaid, or wood cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-636 class 3, PPP-P-585, MIL-B-10377 or PPP-B-601, respectively at the option of the contractor. Shipping containers shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2-inch wide tape conforming to specification PPP-T-76. Boxes may be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes shall be banded with tape conforming to type IV of Specification PPP-T-97 and the appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B.- Regulators, packaged as specified, shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood cleated veneer paper overlaid boxes or class 2 fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377 or PPP-B-636, respectively at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C.- Regulators, packaged as specified, shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Repair parts and accessories.- Repair parts and accessories and special tools (when required) shall be preserved, packaged, and packed for the level of shipment as specified (see 6.1) in accordance with Specification MIL-P-17555.

5.4 Marking.- In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following.

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy-Bureau of Ships
(Project 4220-N019SH)

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